BUS 4028 Assignment - Car Price Dataset

Group 4

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Introduction

This report presents an analysis of a car price dataset using R, focusing on data cleaning, transformation, and exploratory data analysis. The main goal of this study is to prepare the dataset for meaningful analysis and to identify patterns and relationships between key variables that influence car prices.

The dataset contains information on various car attributes such as price, mileage, engine characteristics, and other related features. Before analysis, the data is cleaned to remove missing values and duplicate entries, ensuring accuracy and reliability.

After preprocessing, different statistical and visualization techniques are applied to explore the dataset. These include summary statistics, correlation analysis, and graphical representations to better understand how variables interact with car price.

Overall, this analysis demonstrates how proper data preparation and exploratory techniques can be used to extract useful insights from raw data and support better data-driven decision-making.

The objectives of this analysis include:

1. Print the Structure of your dataset
2. List the variables in your dataset
3. Print the top 15 rows of your dataset
4. Write a user-defined function from dataset
5. Use data manipulation techniques and filter rows
6. Perform data reshaping and joining
7. Remove missing values in your dataset
8. Identify and remove duplicated data in your dataset
9. Reorder multiple rows in (Descending Order)
10. Rename columns names in your dataset
11. Add new variables in your data frame

12. Split dataset into training and testing sets
13. Print the summary statistics of your dataset
14. Perform Mean, Median, Mode and Range with numeric variables
15. Plot a scatter plot of variable in your dataset
16. Plot a bar plot of variables in your dataset
17. Analyze correlation between variables by applying pearson correlation

Load Dataset

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
car_data <- read_csv("car_price_dataset.csv")
```

```
## Rows: 2000 Columns: 11
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): Brand, Fuel_Type, Transmission
## dbl (8): Car_ID, Model_Year, Engine_Size, Mileage, Doors, Owner_Count, Horse...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Print Structure of your Dataset

```r
str(car_data)
```

```
## spc_tbl_ [2,000 x 11] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Car_ID      : num [1:2000] 1 2 3 4 5 6 7 8 9 10 ...
##  $ Brand       : chr [1:2000] "Ford" "Hyundai" "BMW" "Hyundai" ...
##  $ Model_Year  : num [1:2000] 2023 2018 2008 2017 2012 ...
##  $ Engine_Size : num [1:2000] 1.2 3.2 2.2 2.2 2.4 3.5 2.3 3.9 2.6 1.3 ...
##  $ Fuel_Type   : chr [1:2000] "Hybrid" "Electric" "Diesel" "Petrol" ...
##  $ Transmission: chr [1:2000] "Manual" "Manual" "Manual" "Automatic" ...
##  $ Mileage     : num [1:2000] 180635 35628 74672 51246 147233 ...
##  $ Doors       : num [1:2000] 4 2 3 4 3 2 2 4 3 4 ...
##  $ Owner_Count : num [1:2000] 3 4 2 4 4 4 3 1 1 4 ...
##  $ Horsepower  : num [1:2000] 82 259 333 381 290 173 267 369 238 218 ...
##  $ Price       : num [1:2000] 34309 55154 41894 54047 38010 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   Car_ID = col_double(),
##   ..   Brand = col_character(),
##   ..   Model_Year = col_double(),
##   ..   Engine_Size = col_double(),
##   ..   Fuel_Type = col_character(),
##   ..   Transmission = col_character(),
##   ..   Mileage = col_double(),
```

```
##   ..   Doors = col_double(),
##   ..   Owner_Count = col_double(),
##   ..   Horsepower = col_double(),
##   ..   Price = col_double()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

List Variables of the Dataset

```r
head(car_data)
```

```
## # A tibble: 6 x 11
##   Car_ID Brand   Model_Year Engine_Size Fuel_Type Transmission Mileage Doors
##    <dbl> <chr>        <dbl>       <dbl> <chr>     <chr>          <dbl> <dbl>
## 1      1 Ford          2023         1.2 Hybrid    Manual        180635     4
## 2      2 Hyundai       2018         3.2 Electric  Manual         35628     2
## 3      3 BMW           2008         2.2 Diesel    Manual         74672     3
## 4      4 Hyundai       2017         2.2 Petrol    Automatic      51246     4
## 5      5 Hyundai       2012         2.4 Electric  Manual        147233     3
## 6      6 Honda         2023         3.5 Petrol    Automatic      94110     2
## # i 3 more variables: Owner_Count <dbl>, Horsepower <dbl>, Price <dbl>
```

```r
colnames(car_data)
```

```
##  [1] "Car_ID"       "Brand"        "Model_Year"   "Engine_Size"  "Fuel_Type"
##  [6] "Transmission" "Mileage"      "Doors"        "Owner_Count"  "Horsepower"
## [11] "Price"
```

```r
str(car_data)
```

```
## spc_tbl_ [2,000 x 11] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ Car_ID      : num [1:2000] 1 2 3 4 5 6 7 8 9 10 ...
##  $ Brand       : chr [1:2000] "Ford" "Hyundai" "BMW" "Hyundai" ...
##  $ Model_Year  : num [1:2000] 2023 2018 2008 2017 2012 ...
##  $ Engine_Size : num [1:2000] 1.2 3.2 2.2 2.2 2.4 3.5 2.3 3.9 2.6 1.3 ...
##  $ Fuel_Type   : chr [1:2000] "Hybrid" "Electric" "Diesel" "Petrol" ...
##  $ Transmission: chr [1:2000] "Manual" "Manual" "Manual" "Automatic" ...
##  $ Mileage     : num [1:2000] 180635 35628 74672 51246 147233 ...
##  $ Doors       : num [1:2000] 4 2 3 4 3 2 2 4 3 4 ...
##  $ Owner_Count : num [1:2000] 3 4 2 4 4 4 3 1 1 4 ...
##  $ Horsepower  : num [1:2000] 82 259 333 381 290 173 267 369 238 218 ...
##  $ Price       : num [1:2000] 34309 55154 41894 54047 38010 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   Car_ID = col_double(),
##   ..   Brand = col_character(),
##   ..   Model_Year = col_double(),
##   ..   Engine_Size = col_double(),
##   ..   Fuel_Type = col_character(),
##   ..   Transmission = col_character(),
```

```
##   ..   Mileage = col_double(),
##   ..   Doors = col_double(),
##   ..   Owner_Count = col_double(),
##   ..   Horsepower = col_double(),
##   ..   Price = col_double()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

Print the top 15 rows of your dataset

```r
head(car_data, 15)
```

```
## # A tibble: 15 x 11
##    Car_ID Brand   Model_Year Engine_Size Fuel_Type Transmission Mileage Doors
##     <dbl> <chr>        <dbl>       <dbl> <chr>     <chr>          <dbl> <dbl>
##  1      1 Ford          2023         1.2 Hybrid    Manual        180635     4
##  2      2 Hyundai       2018         3.2 Electric  Manual         35628     2
##  3      3 BMW           2008         2.2 Diesel    Manual         74672     3
##  4      4 Hyundai       2017         2.2 Petrol    Automatic      51246     4
##  5      5 Hyundai       2012         2.4 Electric  Manual        147233     3
##  6      6 Honda         2023         3.5 Petrol    Automatic      94110     2
##  7      7 BMW           2016         2.3 Petrol    Manual          7810     2
##  8      8 BMW           2022         3.9 Petrol    Automatic     195932     4
##  9      9 BMW           2010         2.6 Electric  Manual         76688     3
## 10     10 Hyundai       2022         1.3 Electric  Manual         70548     4
## 11     11 Ford          2023         4.1 Petrol    Automatic     198435     2
## 12     12 BMW           2009         2.1 Diesel    Automatic      16162     3
## 13     13 Tesla         2013         2.7 Petrol    Automatic      92858     4
## 14     14 Hyundai       2007         3.7 Diesel    Manual         78535     4
## 15     15 Honda         2014         2.3 Petrol    Automatic     143291     4
## # i 3 more variables: Owner_Count <dbl>, Horsepower <dbl>, Price <dbl>
```

User-Defined Function

```r
calculate_car_value <- function(price, mileage) {
  value <- price - (mileage * 0.05)
  return(value)
}
car_data$Adjusted_Value <- calculate_car_value(
  car_data$Price,
  car_data$Mileage
)

head(car_data)
```

```
## # A tibble: 6 x 12
##   Car_ID Brand   Model_Year Engine_Size Fuel_Type Transmission Mileage Doors
##    <dbl> <chr>        <dbl>       <dbl> <chr>     <chr>          <dbl> <dbl>
## 1      1 Ford          2023         1.2 Hybrid    Manual        180635     4
```

```
## 2      2 Hyundai       2018         3.2 Electric  Manual         35628     2
## 3      3 BMW           2008         2.2 Diesel    Manual         74672     3
## 4      4 Hyundai       2017         2.2 Petrol    Automatic      51246     4
## 5      5 Hyundai       2012         2.4 Electric  Manual        147233     3
## 6      6 Honda         2023         3.5 Petrol    Automatic      94110     2
## # i 4 more variables: Owner_Count <dbl>, Horsepower <dbl>, Price <dbl>,
## #   Adjusted_Value <dbl>
```

Use data manipulation techniques and filter rows

```r
filtered_data <- subset(car_data, Brand == "BMW")

head(filtered_data)
```

```
## # A tibble: 6 x 12
##   Car_ID Brand Model_Year Engine_Size Fuel_Type Transmission Mileage Doors
##    <dbl> <chr>      <dbl>       <dbl> <chr>     <chr>          <dbl> <dbl>
## 1      3 BMW         2008         2.2 Diesel    Manual         74672     3
## 2      7 BMW         2016         2.3 Petrol    Manual          7810     2
## 3      8 BMW         2022         3.9 Petrol    Automatic     195932     4
## 4      9 BMW         2010         2.6 Electric  Manual         76688     3
## 5     12 BMW         2009         2.1 Diesel    Automatic      16162     3
## 6     30 BMW         2008         1.3 Diesel    Automatic      33067     3
## # i 4 more variables: Owner_Count <dbl>, Horsepower <dbl>, Price <dbl>,
## #   Adjusted_Value <dbl>
```

Dependent & Independent Variables, Reshape and Join

```r
dependent_df <- car_data %>%
  select(Car_ID, Price)

independent_df <- car_data %>%
  select(Car_ID, Engine_Size, Horsepower, Mileage, Model_Year, Owner_Count)

independent_long <- independent_df %>%
  pivot_longer(
    cols = -Car_ID,
    names_to = "Variable",
    values_to = "Value"
  )

joined_df <- independent_long %>%
  inner_join(dependent_df, by = "Car_ID")

head(joined_df, 15)
```

```
## # A tibble: 15 x 4
##    Car_ID Variable       Value  Price
##     <dbl> <chr>          <dbl>  <dbl>
```

```
##  1      1 Engine_Size      1.2 34309.
##  2      1 Horsepower      82   34309.
##  3      1 Mileage     180635   34309.
##  4      1 Model_Year    2023   34309.
##  5      1 Owner_Count      3   34309.
##  6      2 Engine_Size      3.2 55154.
##  7      2 Horsepower     259   55154.
##  8      2 Mileage      35628   55154.
##  9      2 Model_Year    2018   55154.
## 10      2 Owner_Count      4   55154.
## 11      3 Engine_Size      2.2 41894.
## 12      3 Horsepower     333   41894.
## 13      3 Mileage      74672   41894.
## 14      3 Model_Year    2008   41894.
## 15      3 Owner_Count      2   41894.
```

Data Cleaning

Remove Missing Values

```r
clean_data <- na.omit(car_data)
head(clean_data)
```

```
## # A tibble: 6 x 12
##   Car_ID Brand   Model_Year Engine_Size Fuel_Type Transmission Mileage Doors
##    <dbl> <chr>        <dbl>       <dbl> <chr>     <chr>          <dbl> <dbl>
## 1      1 Ford          2023         1.2 Hybrid    Manual        180635     4
## 2      2 Hyundai       2018         3.2 Electric  Manual         35628     2
## 3      3 BMW           2008         2.2 Diesel    Manual         74672     3
## 4      4 Hyundai       2017         2.2 Petrol    Automatic      51246     4
## 5      5 Hyundai       2012         2.4 Electric  Manual        147233     3
## 6      6 Honda         2023         3.5 Petrol    Automatic      94110     2
## # i 4 more variables: Owner_Count <dbl>, Horsepower <dbl>, Price <dbl>,
## #   Adjusted_Value <dbl>
```

Remove Duplicate Records

```r
clean_data <- distinct(clean_data)
head(clean_data)
```

```
## # A tibble: 6 x 12
##   Car_ID Brand   Model_Year Engine_Size Fuel_Type Transmission Mileage Doors
##    <dbl> <chr>        <dbl>       <dbl> <chr>     <chr>          <dbl> <dbl>
## 1      1 Ford          2023         1.2 Hybrid    Manual        180635     4
## 2      2 Hyundai       2018         3.2 Electric  Manual         35628     2
## 3      3 BMW           2008         2.2 Diesel    Manual         74672     3
## 4      4 Hyundai       2017         2.2 Petrol    Automatic      51246     4
## 5      5 Hyundai       2012         2.4 Electric  Manual        147233     3
## 6      6 Honda         2023         3.5 Petrol    Automatic      94110     2
## # i 4 more variables: Owner_Count <dbl>, Horsepower <dbl>, Price <dbl>,
## #   Adjusted_Value <dbl>
```

Sort Data by Car Price (Descending)

```r
colnames(clean_data)
```

```
##  [1] "Car_ID"         "Brand"          "Model_Year"     "Engine_Size"
##  [5] "Fuel_Type"      "Transmission"   "Mileage"        "Doors"
##  [9] "Owner_Count"    "Horsepower"     "Price"          "Adjusted_Value"
```

```r
sorted_data <- clean_data %>%
  arrange(desc(Price))

head(sorted_data)
```

```
## # A tibble: 6 x 12
##   Car_ID Brand   Model_Year Engine_Size Fuel_Type Transmission Mileage Doors
##    <dbl> <chr>        <dbl>       <dbl> <chr>     <chr>          <dbl> <dbl>
## 1   1963 Hyundai       2018         4.6 Petrol    Automatic      13744     3
## 2    150 Tesla         2022         4.3 Diesel    Automatic      27228     3
## 3    743 Ford          2023         4.6 Petrol    Manual         60401     4
## 4    916 Honda         2013         5   Petrol    Manual         20588     2
## 5   1709 Honda         2022         5   Electric  Automatic     114034     4
## 6   1724 Honda         2022         3.9 Hybrid    Manual         13720     2
## # i 4 more variables: Owner_Count <dbl>, Horsepower <dbl>, Price <dbl>,
## #   Adjusted_Value <dbl>
```

Rename Columns names

```r
clean_data <- clean_data %>%
  rename(
    Car_Price = Price,
    Car_Mileage = Mileage
  )

head(clean_data)
```

```
## # A tibble: 6 x 12
##   Car_ID Brand   Model_Year Engine_Size Fuel_Type Transmission Car_Mileage Doors
##    <dbl> <chr>        <dbl>       <dbl> <chr>     <chr>              <dbl> <dbl>
## 1      1 Ford          2023         1.2 Hybrid    Manual            180635     4
## 2      2 Hyundai       2018         3.2 Electric  Manual             35628     2
## 3      3 BMW           2008         2.2 Diesel    Manual             74672     3
## 4      4 Hyundai       2017         2.2 Petrol    Automatic          51246     4
## 5      5 Hyundai       2012         2.4 Electric  Manual            147233     3
## 6      6 Honda         2023         3.5 Petrol    Automatic          94110     2
## # i 4 more variables: Owner_Count <dbl>, Horsepower <dbl>, Car_Price <dbl>,
## #   Adjusted_Value <dbl>
```

Add New Variables in data frame

```r
car_data <- car_data %>%
  mutate(
    Price_Per_Horsepower = round(Price / Horsepower, 2),
    Engine_Size_CC = Engine_Size * 1000
  )

car_data %>%
  select(Brand, Price, Horsepower, Price_Per_Horsepower,
         Engine_Size, Engine_Size_CC) %>%
  head(10)
```

```
## # A tibble: 10 x 6
##    Brand    Price Horsepower Price_Per_Horsepower Engine_Size Engine_Size_CC
##    <chr>    <dbl>      <dbl>                <dbl>       <dbl>          <dbl>
##  1 Ford    34309.         82                 418.         1.2           1200
##  2 Hyundai 55154.        259                 213.         3.2           3200
##  3 BMW     41894.        333                 126.         2.2           2200
##  4 Hyundai 54047.        381                 142.         2.2           2200
##  5 Hyundai 38010.        290                 131.         2.4           2400
##  6 Honda   54182.        173                 313.         3.5           3500
##  7 BMW     52140.        267                 195.         2.3           2300
##  8 BMW     61707.        369                 167.         3.9           3900
##  9 BMW     44993.        238                 189.         2.6           2600
## 10 Hyundai 40925.        218                 188.         1.3           1300
```

Create Training and Test Set

```r
set.seed(123)

train_indices <- sample(
  1:nrow(car_data),
  size = floor(0.80 * nrow(car_data)),
  replace = FALSE
)

train_set <- car_data %>% slice(train_indices)
test_set <- car_data %>% slice(-train_indices)

cat("Total rows:        ", nrow(car_data), "\n")
```

```
## Total rows:         2000
```

```r
cat("Training set rows: ", nrow(train_set), "\n")
```

```
## Training set rows:  1600
```

```r
cat("Test set rows:     ", nrow(test_set), "\n")
```

```
## Test set rows:      400
```

Summary Statistics

```r
summary(car_data)
```

```
##      Car_ID          Brand             Model_Year    Engine_Size
##  Min.   :   1.0   Length:2000        Min.   :2005   Min.   :1.000
##  1st Qu.: 500.8   Class :character   1st Qu.:2009   1st Qu.:2.000
##  Median :1000.5   Mode  :character   Median :2014   Median :2.900
##  Mean   :1000.5                      Mean   :2014   Mean   :2.968
##  3rd Qu.:1500.2                      3rd Qu.:2019   3rd Qu.:4.000
##  Max.   :2000.0                      Max.   :2023   Max.   :5.000
##   Fuel_Type         Transmission          Mileage           Doors
##  Length:2000        Length:2000        Min.   :  5036   Min.   :2.000
##  Class :character   Class :character   1st Qu.: 52366   1st Qu.:2.000
##  Mode  :character   Mode  :character   Median :100591   Median :3.000
##                                        Mean   :100737   Mean   :2.988
##                                        3rd Qu.:148025   3rd Qu.:4.000
##                                        Max.   :199904   Max.   :4.000
##   Owner_Count      Horsepower        Price       Adjusted_Value
##  Min.   :1.000   Min.   : 70.0   Min.   :18912   Min.   :10668
##  1st Qu.:2.000   1st Qu.:154.0   1st Qu.:39764   1st Qu.:34237
##  Median :3.000   Median :236.0   Median :46112   Median :40875
##  Mean   :2.533   Mean   :235.7   Mean   :46170   Mean   :41133
##  3rd Qu.:4.000   3rd Qu.:319.0   3rd Qu.:52471   3rd Qu.:48346
##  Max.   :4.000   Max.   :399.0   Max.   :72268   Max.   :71581
##  Price_Per_Horsepower Engine_Size_CC
##  Min.   : 85.39       Min.   :1000
##  1st Qu.:157.39       1st Qu.:2000
##  Median :194.72       Median :2900
##  Mean   :232.06       Mean   :2968
##  3rd Qu.:274.33       3rd Qu.:4000
##  Max.   :782.68       Max.   :5000
```

Mean, Median, Mode and Range

```r
# Function to calculate mode
get_mode <- function(v) {
  uniqv <- unique(v)
  uniqv[which.max(tabulate(match(v, uniqv)))]
}

# Select Price column
col_target <- car_data$Price

# Calculate statistics
```

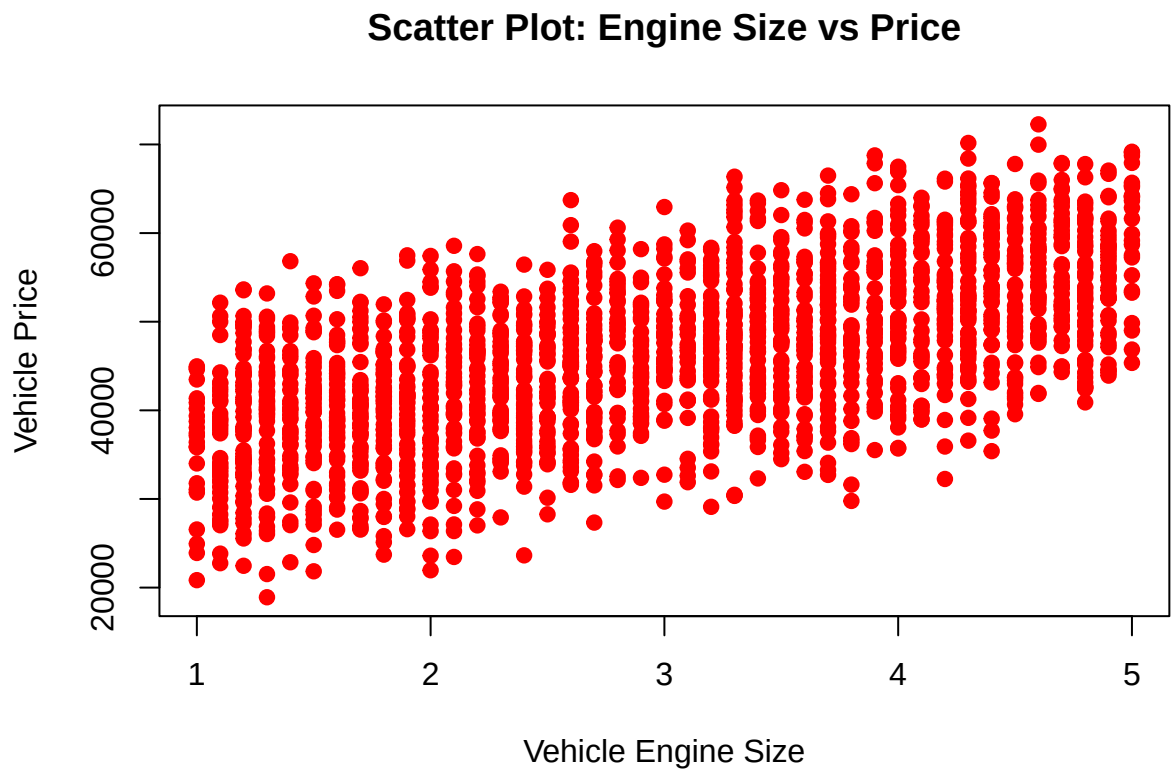
```r
Statistics_Summary <- list(
  Mean = mean(col_target, na.rm = TRUE),
  Median = median(col_target, na.rm = TRUE),
  Mode = get_mode(col_target),
  Standard_Deviation = sd(col_target, na.rm = TRUE),
  Min = min(col_target, na.rm = TRUE),
  Max = max(col_target, na.rm = TRUE),
  Range = max(col_target, na.rm = TRUE) - min(col_target, na.rm = TRUE)
)

Statistics_Summary
```

```
## $Mean
## [1] 46169.51
##
## $Median
## [1] 46112.35
##
## $Mode
## [1] 51222.4
##
## $Standard_Deviation
## [1] 9211.686
##
## $Min
## [1] 18911.55
##
## $Max
## [1] 72267.8
##
## $Range
## [1] 53356.25
```

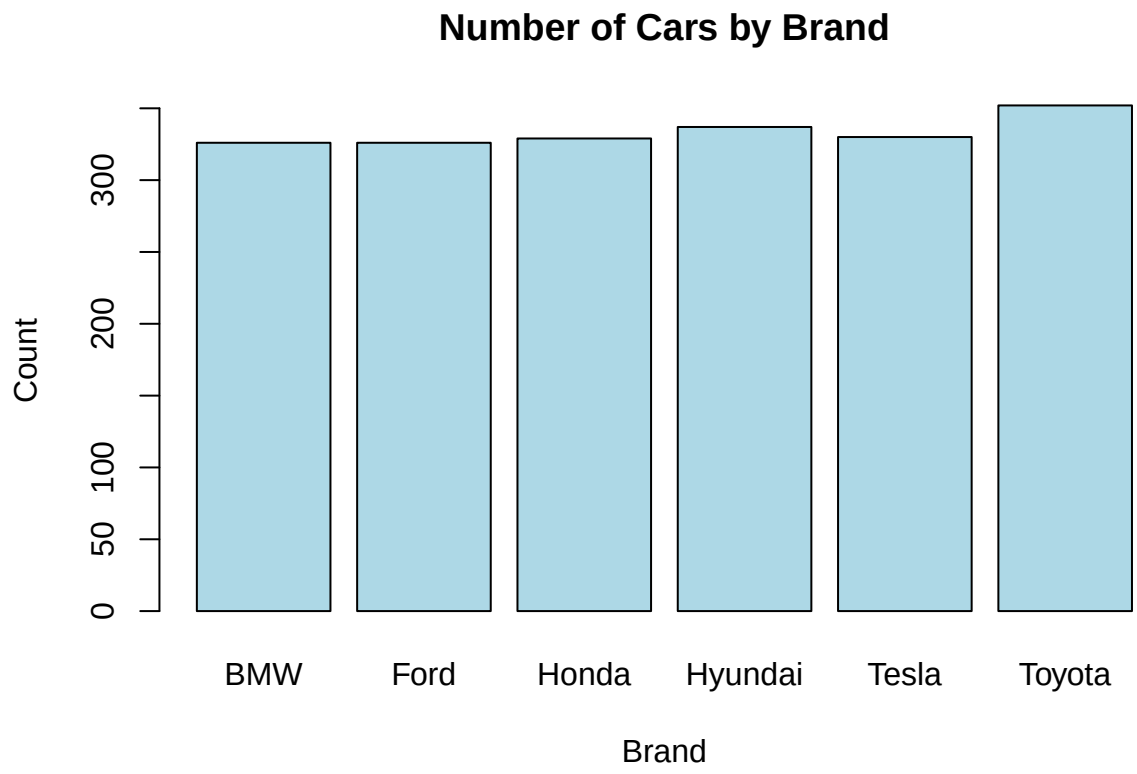
Scatter Plot of variables

```r
plot(car_data$Engine_Size, car_data$Price,
     main = "Scatter Plot: Engine Size vs Price",
     xlab = "Vehicle Engine Size",
     ylab = "Vehicle Price",
     pch = 19,
     col = "red")
```



Bar plot of car brands

```
barplot(table(car_data$Brand),
        main = "Number of Cars by Brand",
        xlab = "Brand",
        ylab = "Count",
        col = "lightblue")
```



Pearson Correlation

```
correlation_result <- cor(
  car_data$Price,
  car_data$Horsepower,
  method = "pearson"
)

cat("Pearson Correlation between Price and Horsepower:",
    round(correlation_result, 4), "\n")
```

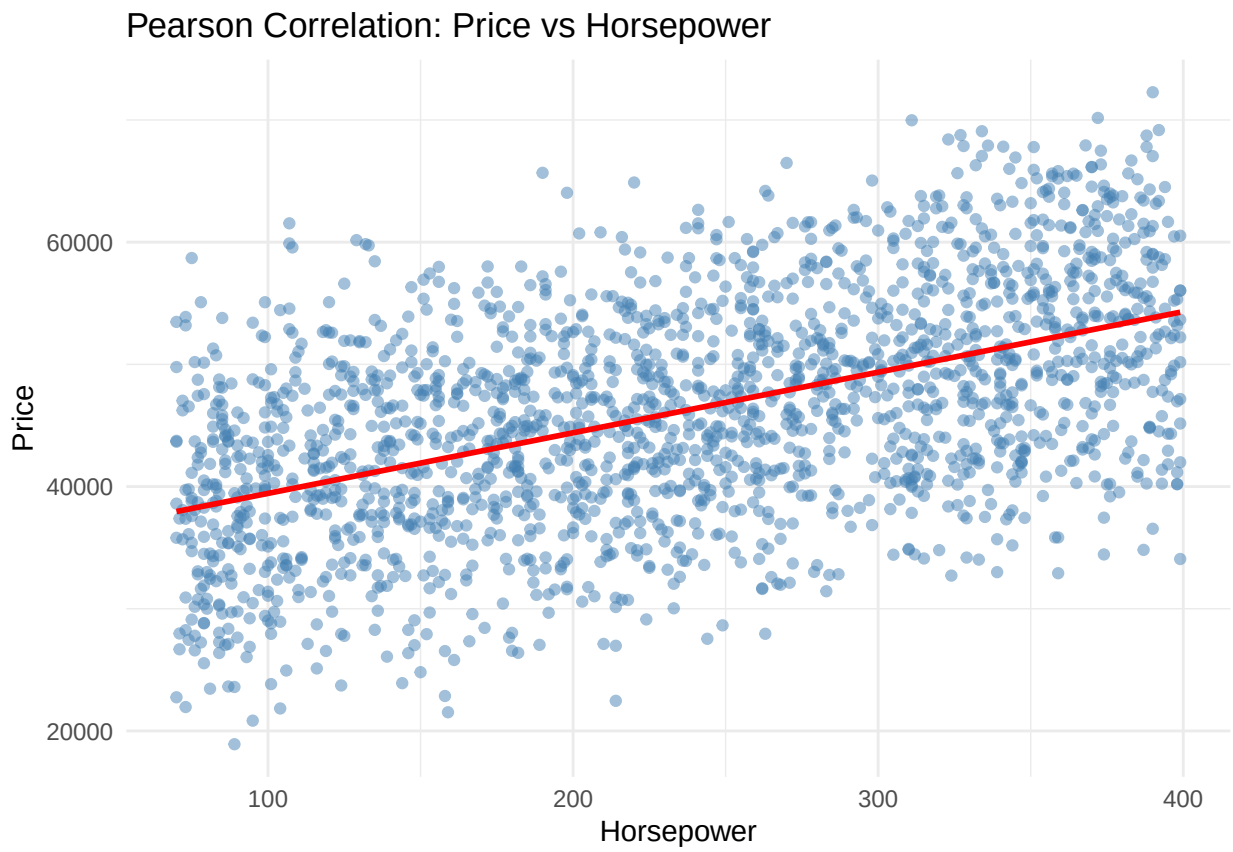
```
## Pearson Correlation between Price and Horsepower: 0.5146
```

Visualization

```
ggplot(car_data, aes(x = Horsepower, y = Price)) +
  geom_point(color = "steelblue", alpha = 0.5) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(
    title = "Pearson Correlation: Price vs Horsepower",
    x = "Horsepower",
```

```
    y = "Price"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Conclusion

his study focused on the cleaning, transformation, and analysis of a car price dataset using R. The dataset was first explored to understand its structure and variables, followed by essential data preparation steps such as handling missing values, removing duplicate records, and organizing the data for analysis.

After cleaning, various statistical and visual analysis techniques were applied to gain insights into the relationships between key variables such as price, mileage, and other vehicle attributes. Summary statistics and correlation analysis helped identify patterns and potential relationships within the dataset, while visualizations provided a clearer understanding of trends and distributions.

Overall, the analysis demonstrates the importance of proper data preparation before performing any meaningful statistical analysis. Clean and well-structured data improves the reliability of results and supports better decision-making. Future analysis could involve building predictive models to estimate car prices based on selected features.